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Eliminating Deburring Run in Multistage Single Trip Completions Improving Efficiency and Reducing Costs

F. Ortega, M. Dunn, M. Ischy, G. Corbett, L. Izquierdo, and M. Metzmaier, Chevron U.S.A. Inc., Houston, TX, USA;
Y. Esquitin and N. Sperle, Welltec Inc. Katy, TX, USA

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Abstract

Operational efficiency is critical to the economic viability of deepwater wells. To accelerate completion timelines, the Multistage Single Trip (MSST) system was introduced in the early 2000s, enabling frac-and-pack stimulation across multiple intervals in a single deployment. While this innovation significantly reduced rig time and operational complexity, it necessitated a dedicated deburring run to mitigate the risk of damage to isolation packers caused by perforation burrs.

To further optimize the MSST system, metal expandable technology was integrated, eliminating the need for the deburring trip. This paper presents the planning, testing, qualification, deployment, and field results of incorporating the Well Expandable Inner String Packer (WEA-ISP)—comprising the Metal Expandable Packer (MEP) and Well Expandable Anchor (WEA)—into the MSST system. This integration replaced legacy isolation packers and enabled safe passage through perforated intervals without compromising zonal isolation or system integrity.

A multidisciplinary collaboration ensured the successful deployment of WEA-ISP without altering well objectives. Key design parameters—including burr height, perforation length, and clearance—were identified to perform rigorous test on the WEA-ISP. The system underwent comprehensive ISO 14310 qualification, demonstrating its ability to withstand up to 15,000 psi differential pressure and anchoring forces exceeding 480,000 lbf.

Field deployments in four deepwater wells, including both water injectors and oil producers in the Gulf of America, validated the system's performance. The integration of WEA-ISP into the MSST system eliminated the deburring run, saving an average of 2.7 days of rig time per ~30,000 ft well—equating to over \$3 million in cost savings per well. The system supports both 10,000 psi and 15,000 psi treatment pressures without requiring additional inventory.

This paper provides detailed insights into qualification protocols, engineering documentation, operational procedures, and contingency planning. The successful implementation of WEA-ISP within MSST completions demonstrates a robust, cost-effective solution that maintains reliable zonal isolation while significantly enhancing operational efficiency.

Introduction

The Wilcox Formation in the Gulf of America (GoA) presents a complex subsurface environment characterized by interbedded sandstones, shales, and lignite deposits. These heterogeneous lithologies, formed through fluvial, deltaic, and shallow marine depositional processes, pose significant challenges for well completions and stimulation. Advanced technologies are required to ensure effective zonal isolation and stimulation across multiple intervals.

To address these challenges, the Multistage Single Trip (MSST) system has been widely adopted in GoA deepwater developments. The MSST system enables frac-and-pack stimulation across multiple zones in a single deployment, significantly reducing rig time and operational costs. However, conventional MSST operations often require a dedicated deburring run to remove perforation burrs, which can damage isolation packers and compromise system integrity.

Table 1 outlines the typical rig-time allocation for a three-zone MSST deployment in a 28,000 ft well prior to stimulation operations:

Table 1—Conventional MSST deployment steps and rig time.

Step	Task description	Rig time (days)
1	Set sump packer on wireline.	0.5
2	Perforate well with Tubing Conveyed Perforating (TCP)	2.5
3	Deburr perforations	2.7
4	Run MSST assembly and seal in sump packer	2.5
	Total time	8.2

To further enhance operational efficiency, the integration of Metal Expandable Packer (MEP) technology was evaluated. Specifically, the Well Expandable Anchor–Inner String Packer (WEA-ISP) system was proposed to replace legacy hydraulically set mechanical packers. If proven successful, this integration would eliminate the need for a deburring run, potentially saving 2.5–3 days of rig time per well.

A rigorous testing and qualification program was initiated to validate the WEA-ISP's performance under simulated downhole conditions. This included mechanical integrity testing through burr-emulating fixtures and full ISO 14310 qualification for sealing and anchoring capabilities. The objective was to ensure the WEA-ISP could reliably pass through perforation burrs and maintain zonal isolation integrity during MSST completions.

Objective and Collaborative Development Approach

The primary objective of this paper is to present the methodology and results of integrating the Well Expandable Anchor–Inner String Packer (WEA-ISP) into Multistage Single Trip (MSST) completion designs, with the aim of enhancing operational efficiency in deepwater wells. This integration seeks to eliminate the need for a dedicated deburring run, thereby reducing rig time and associated costs while maintaining reliable zonal isolation.

The paper outlines the comprehensive testing and qualification protocols undertaken to validate the WEA-ISP system, including mechanical integrity assessments, ISO 14310 certification, and compatibility evaluations with existing MSST infrastructure. Additionally, it details the job planning, risk mitigation strategies, and operational benefits realized through field deployments.

To ensure a robust and technically sound implementation, a multidisciplinary team was assembled in MSST technology, Tubing Conveyed Perforating (TCP), WEA-ISP design, and completions engineering. This collaborative group was tasked with:

- Defining the WEA-ISP system parameters specific to deepwater GoA wells.
- Developing a testing protocol to simulate perforation burrs and assess system durability.
- Establishing qualification criteria and procedures to evaluate sealing and anchoring performance under representative downhole conditions.

This paper documents the systematic approach taken to validate the WEA-ISP as a viable enhancement to MSST completions, culminating in four successful deployments to date in the Gulf of America.

WEA-ISP adoption methodology

A value-driven engineering approach was adopted to evaluate the feasibility of incorporating the WEA-ISP into the MSST system. Drawing from prior successful deployments, the proposed solution (Fig. 1) was presented to the completions engineering team of a major deepwater GoA project. The integration aimed to replace legacy isolation packers, eliminate the deburring run, and deliver significant rig-time savings.

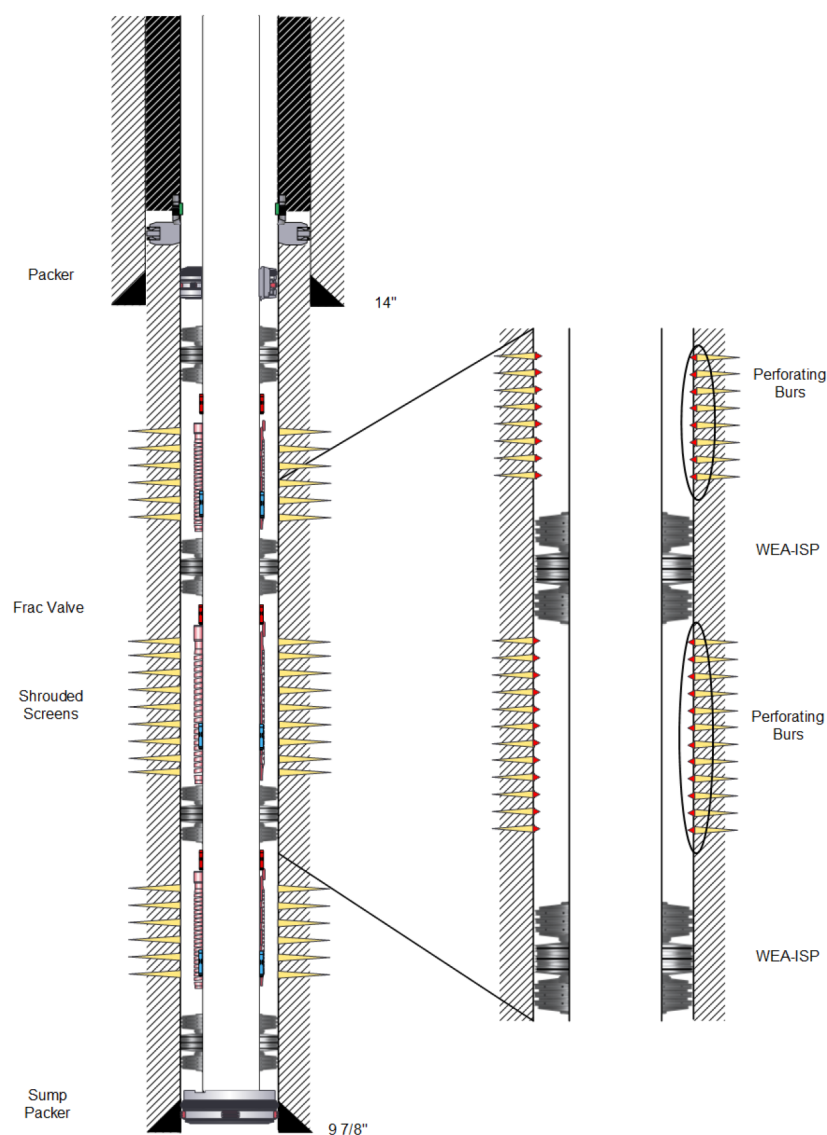


Figure 1—WEA-ISP incorporation to the MSST system, no deburring run performed.

Following extensive technical reviews and stakeholder engagement, the first trial deployment was approved upon completion of the following milestones:

1. Feasibility Assessment:

- Failure Mode, Effects, and Criticality Analysis (FMECA)
- Finite Element Analysis (FEA) modeling
- Burr damage simulation using a test fixture
- ISO 14310 qualification for anchoring and sealing

2. System Compatibility Review:

- Surface integration testing at the MSST system provider's facility

3. Operational Planning:

- Detailed job planning and Job Risk Assessment (JRA)

WEA-ISP System Requirements

The design and qualification of the WEA-ISP system began with a comprehensive Statement of Requirements (SOR), capturing all critical downhole conditions and life-of-well expectations. The selected WEA-ISP configuration was tailored to meet the specific demands of the GoA deepwater environment.

Table 2—WEA-ISP Equipment requirements for the deepwater well in GoA.

Expected work life of the WEA-ISP	25 years of productive life
Qualification standard	ISO14310 V3
Base pipe	6 5/8"
Maximum bur height to pass through	0.31"
Maximum Setting ID	9"
Running outer diameter	8.2"
Maximum expansion pressure	5,000 psi
Annulus differential pressure	15,000 psi from above or below
Anchoring requirement	450,000 lbs
Maximum temperature	265 DegF
Minimum temperature	122 DegF
Metallurgy	Corrosive resistant alloy – Nickel and Chrome
Torque, tensile and compression limits	Retain completion specifications

Material and Component Design

Metallurgy Selection. The metallurgy was selected to withstand long-term exposure to corrosive production environments, ensuring integrity over the 25-year life of the well.

Seal Assembly Design. The MEP features a pressure-activated expansion sleeve. Upon expansion, the elastomer seals are energized to create metal-to-metal and elastomer-to-metal contact. Initial contact stress ensures sealing at low pressures, while increasing annular pressure further energizes the seal against the formation.

Anchoring Assembly Design. The anchoring mechanism comprises multiple slip segments distributed across dual expansion sleeves. These slips grip the parent casing without causing damage, delivering anchoring forces exceeding 480,000 lbf.

The final WEA-ISP configuration was engineered to meet the high-pressure and high-temperature demands of GoA deepwater wells, ensuring robust performance and reliable zonal isolation.

WEA-ISP Components and Actuation Mechanism. The Well Expandable Anchor–Inner String Packer (WEA-ISP) is an annular casing-mounted system designed to provide reliable zonal isolation and anchoring in deepwater completions. It consists of a packer and anchor assembly integrated onto the outer surface of a base pipe. The system is engineered to expand plastically and conform to the wellbore and casing geometry, eliminating the need for post-perforation deburring operations.

Figure 2 illustrates the key components of the WEA-ISP system, which are described below:

1. **BasePipe** - Serves as the structural backbone of the system, providing mechanical integrity and fluid conveyance.
2. **End Piece** - Provides structural robustness and pressure containment. Typically manufactured from high-strength corrosion-resistant alloys such as Inconel 718 or Super Duplex stainless steel.
3. **Packer Expansion Sleeve** - A deformable metal sleeve designed to plastically yield under applied pressure, allowing it to conform to the wellbore geometry and initiate sealing.
4. **Seal Assembly** - Composed of elastomeric elements mounted externally on the expansion sleeve. These seals are energized during expansion to establish a reliable barrier against the formation, ensuring zonal isolation.
5. **Anchor Expansion Sleeve** - Similar in design to the packer sleeve, this component plastically deforms to engage the parent casing, enabling anchoring of the system.
6. **Bi-Directional Anchors** - A series of slip segments engineered to grip the casing in both axial directions. These slips prevent movement of the WEA-ISP system during stimulation and production operations, ensuring mechanical stability.

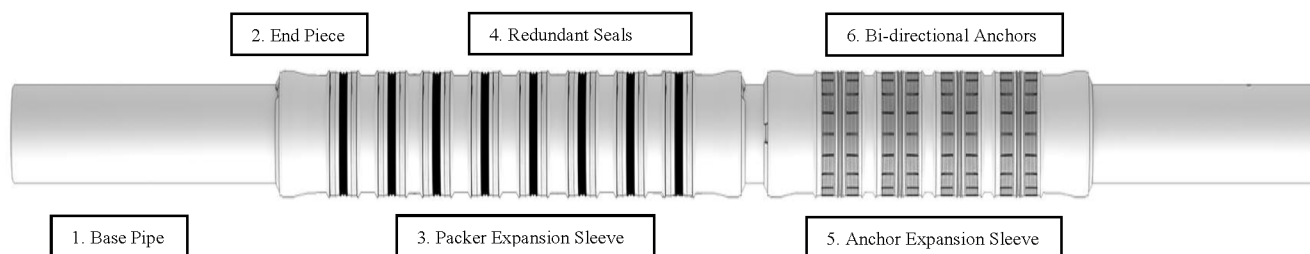


Figure 2—WEA-ISP components

The actuation of the WEA-ISP is purely pressure-driven. Upon reaching the designed expansion pressure, the sleeves plastically deform, activating the seal and anchor assemblies. The system is designed to withstand high differential pressures and axial loads, making it suitable for demanding deepwater environments.

WEA-ISP Testing and Qualification Philosophy

To validate the robustness of the WEA-ISP system under realistic downhole conditions, a comprehensive testing and qualification program was developed. The primary objective was to simulate the mechanical damage that could occur during passage through perforated casing—specifically burrs generated by high-density perforating guns—and to ensure the system could maintain integrity and functionality post-damage.

Perforation Burr Characterization. A certified evaluation was conducted using a 6.62" High Pressure High Shot Density (HP-HSD) perforating system (18 shots per foot) in a 9 7/8" casing, following API RP 19B Section 1 guidelines. The resulting burr profile was documented, with a maximum burr height of 0.25" and an average of 0.13", as shown in Figure 3.

Shot No.	No 1	No 2	No 3	No 4	No 5	No 6	No 7	No 8	No 9	No 10	No 11
Clearance, in.	0.62	1.18	1.18	0.80	1.38	0.80	0.62	1.18	1.18	0.80	1.38
Casing Hole Diameter, Short Axis, in.	0.93	0.87	0.77	0.86	0.56	0.81	0.88	0.74	0.46	0.87	0.70
Casing Hole Diameter, Long Axis, in.	0.93	0.93	0.91	0.98	0.93	0.99	0.96	0.82	1.02	1.04	0.77
Average Casing Hole Diameter, in.	0.93	0.90	0.84	0.92	0.75	0.90	0.92	0.78	0.74	0.96	0.74
Total Depth, in.	7.0	7.6	6.4	6.8	5.8	6.2	6.3	11.2	5.6	5.8	6.1
Burr Height, in.	0.17	0.20	0.07	0.13	0.07	0.10	0.09	0.09	0.08	0.11	0.12

Shot No.	No 12	No 13	No 14	No 15	No 16	No 17	No 18	No 19	No 20	No 21	No 22
Clearance, in.	0.80	0.62	1.18	1.18	0.80	1.38	0.80	0.62	1.18	1.18	0.80
Casing Hole Diameter, Short Axis, in.	0.83	0.94	0.82	0.65	0.92	0.67	0.79	1.02	0.63	0.77	0.80
Casing Hole Diameter, Long Axis, in.	0.93	0.97	0.88	0.94	0.97	0.98	0.86	1.01	0.83	0.86	0.95
Average Casing Hole Diameter, in.	0.88	0.96	0.85	0.80	0.95	0.83	0.83	1.02	0.73	0.82	0.88
Total Depth, in.	5.7	4.6	5.6	6.6	5.8	5.9	7.3	6.8	5.4	6.1	6.3
Burr Height, in.	0.18	0.25	0.12	0.07	0.07	0.12	0.08	0.10	0.06	0.12	0.16

Shot No.	No 23	No 24	No 25	No 26	No 27	No 28	No 29	No 30	No 31	No 32	No 33
Clearance, in.	1.29	0.78	0.63	1.11	1.11	0.78	1.29	0.78	0.63	1.11	1.11
Casing Hole Diameter, Short Axis, in.	0.40	0.91	0.84	0.64	0.43	0.82	0.58	0.87	0.84	0.71	0.78
Casing Hole Diameter, Long Axis, in.	0.50	0.95	0.95	0.71	0.56	0.85	0.62	0.88	0.94	0.81	0.88
Average Casing Hole Diameter, in.	0.45	0.93	0.90	0.68	0.50	0.84	0.60	0.88	0.89	0.76	0.83
Total Depth, in.	5.6	5.6	5.9	5.0	4.8	6.8	5.4	5.8	7.7	4.8	6.2
Burr Height, in.	0.13	0.13	0.13	0.06	0.16	0.08	0.08	0.05	0.23	0.18	0.19

Shot No.	No 34	No 35	No 36	No 37	No 38	No 39	No 40	No 41	No 42	No 43	No 44	Average
Clearance, in.	0.78	1.29	0.78									0.72
Casing Hole Diameter, Short Axis, in.	0.81	0.48	0.66									0.82
Casing Hole Diameter, Long Axis, in.	0.87	0.58	0.71									0.77
Average Casing Hole Diameter, in.	0.84	0.53	0.69									0.77
Total Depth, in.	8.2	4.9	7.3									6.5
Burr Height, in.	0.11	0.10	0.20									0.13

Figure 3—Maximum perforation burr height from testing certified by API

FEA Modeling of Seal Performance. Using the burr data, a detailed Finite Element Analysis (FEA) was performed to assess the impact of induced damage on the sealing performance of the WEA-ISP. The model simulated multiple redundant seals encountering burrs at varying angles and depths. Figure 4 illustrates the seal deformation and pressure conformance under these conditions.

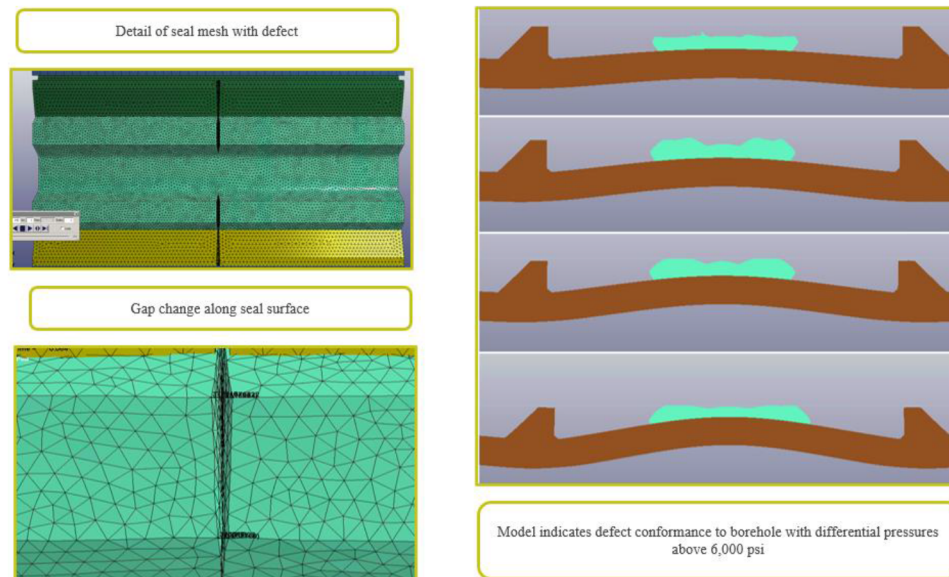


Figure 4—FEA evaluation of the induced damaged on the seals and the sealing conformance with different pressures

Physical Damage Simulation. Two physical test setups were designed to emulate burr-induced damage:

- **Setup 1:** A controlled fixture with calibrated burr height and angle was used to induce damage across all seal elements of the WEA-ISP assembly (Figure 5).

- **Setup 2:** A test vessel containing multiple sharp tips at varied angles and heights was engineered to simulate perforation burrs. The WEA-ISP was passed repeatedly through the vessel to generate realistic wear and tear on its components (Figures 6 & 7).

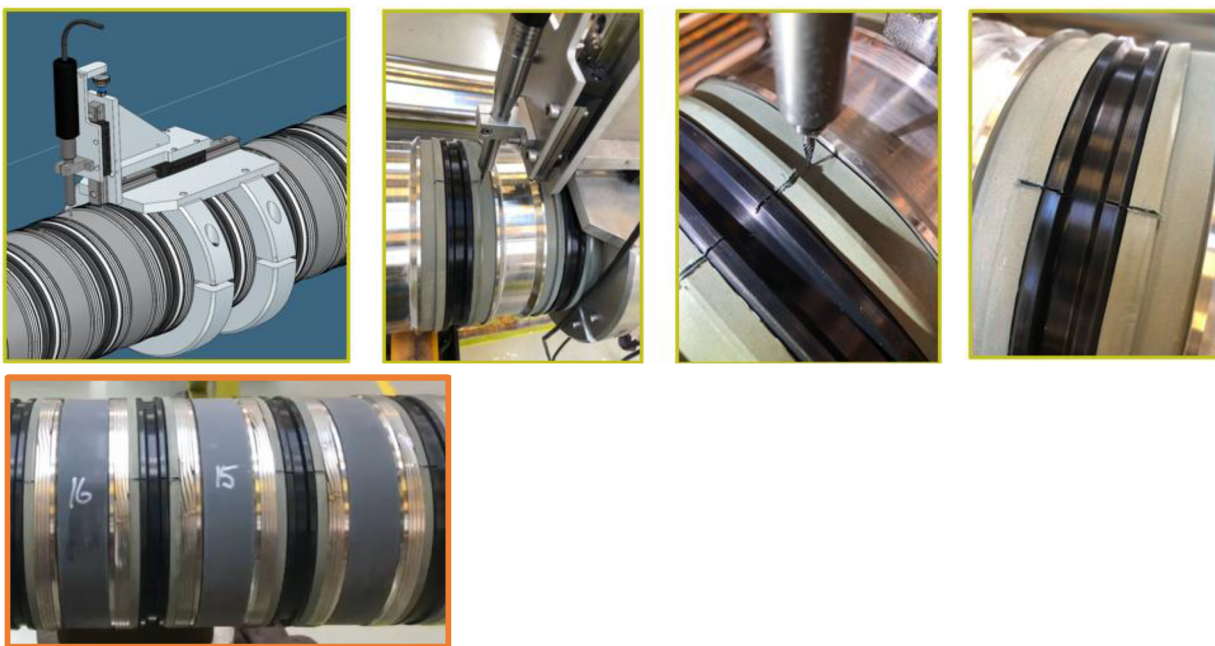


Figure 5—Induced damage to the seal with specific depth of penetration and angle in each redundant seal across the entire WEA-ISP assembly



Figure 6—Test setup for passing the WEA-ISP assembly through emulated burrs

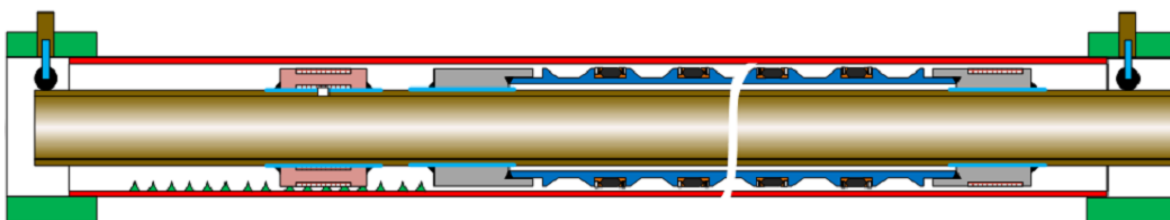


Figure 7—Showing the schematic of the passing the WEA-ISP system through the setup with sharp tips to emulate the burrs.

Post-testing inspections revealed superficial damage to components such as the end piece, expansion sleeves, seals, and cover plates (Figures 8 & 9). However, no critical damage was observed that would compromise system integrity or performance.

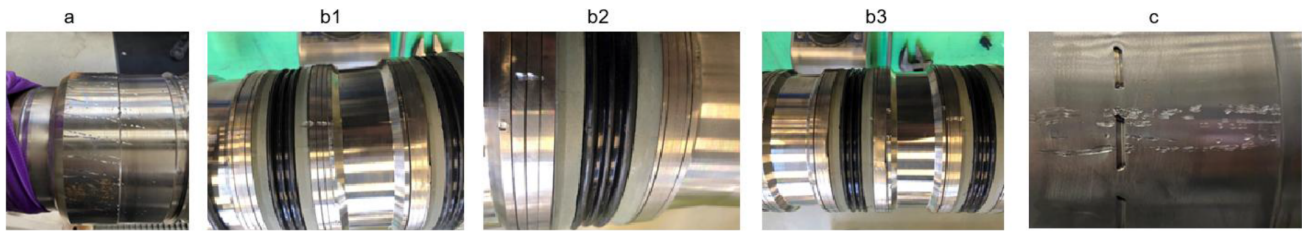


Figure 8—a) damage to end piece, b1, b2 and b3) damage expansion sleeve and seals, c) damage to cover plate

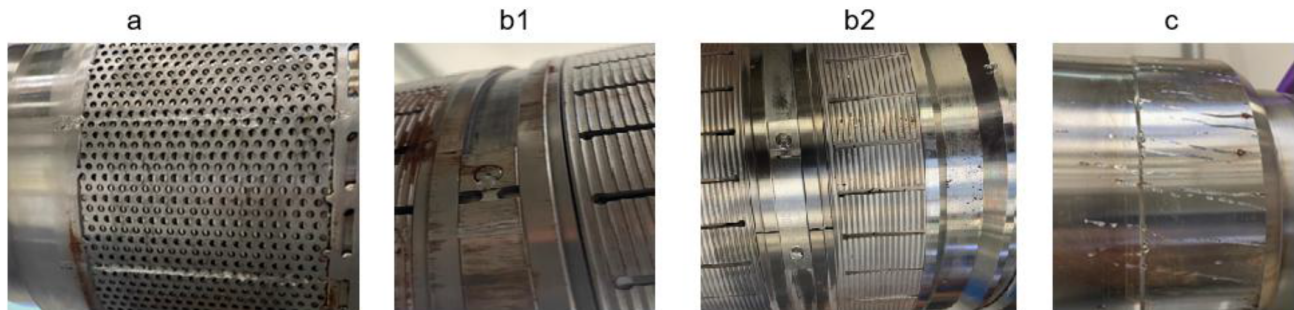


Figure 9—a) damage to filter cover, b1 and b2) damage expansion sleeve and slips, c) damage to end piece

Qualification Process. Following damage simulation, the WEA-ISP underwent full ISO 14310 V3 qualification, designed to validate sealing and anchoring performance under extreme conditions:

- **Test Conditions:**

- Differential pressure: 15,000 psi
- Temperature range: 212°F delta
- Axial load: up to 480,000 lbf
- Medium: Water
- Orientation: Horizontal and centralized

- **Instrumentation:**

- Pressure zones: PC (internal sleeve), PA/PB (annulus), PD (bore)
- Monitoring: Electronic pressure gauges, strain gauges, temperature sensors
- Heating: External blankets applied to simulate thermal cycling (Figures 10–12)

Axial load testing was conducted by fixing the outer test pipe and applying force to the inner liner. Movement was tracked using rope sensors, and net load calculations were used to define the packer's load envelope.

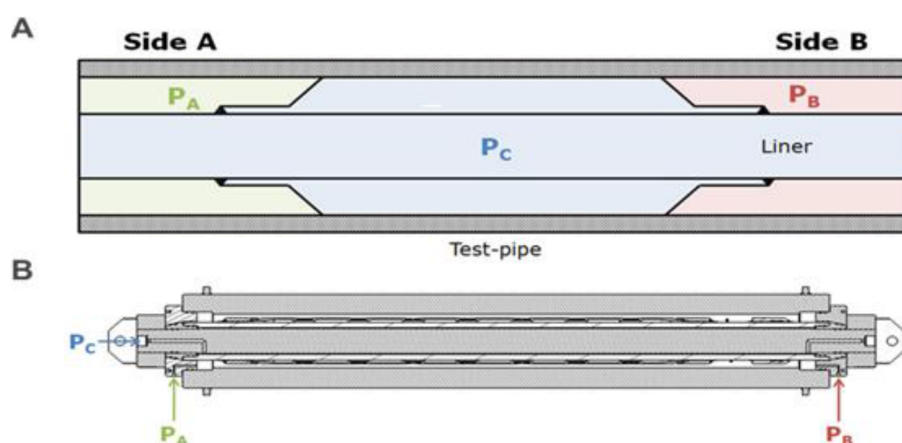


Figure 10—Pressure test setup for ISO14310 qualification

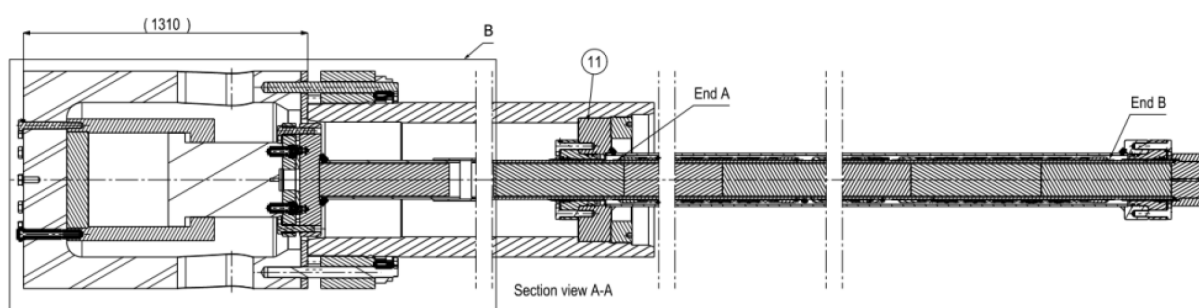


Figure 11—Test setup for ISO14310 with pull/push test jig attached

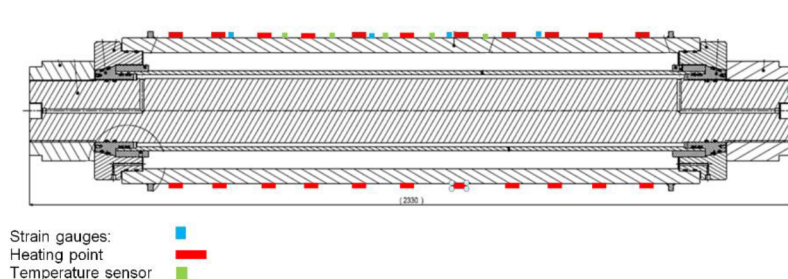


Figure 12—a) Strain gauges position and temperature sensors. b) heating blanket setup

Test Results and Qualification Outcome. The WEA-ISP successfully met all ISO 14310 V3 qualification criteria, including:

- **Sealing Integrity:** Maintained across full pressure reversals and temperature cycles, with less than 1% reduction in rated differential pressure.
- **Anchoring Stability:** No axial movement observed during hold periods under maximum load.
- **Damage Tolerance:** Despite induced damage to the seals, the system demonstrated full functionality and no signs of compromised integrity.

These results confirm the WEA-ISP's suitability for deployment in deepwater MSST completions, offering a robust solution capable of withstanding harsh downhole environments and eliminating the need for a deburring run.

WEA-ISP adoption and job execution

Following the successful qualification of the WEA-ISP system demonstrated through repeated passes across emulated perforation burrs and post-damage integrity testing, a candidate well was selected for field deployment. A comprehensive Failure Mode, Effects, and Criticality Analysis (FMECA), along with detailed job planning and execution protocols, was developed to ensure a successful trial run.

Risk Assessment. A thorough FMECA was conducted by the multidisciplinary team to evaluate potential risks and ensure seamless integration of the WEA-ISP into the MSST system. The analysis covered all operational phases, including pick-up (PU), run-in-hole (RIH), and expansion. Figure 13 presents a section of the FMECA focused on PU and RIH operations.

3	Pick-Up(PU) and Run-in-hole (RIH) with MSST System							
3.1	Perform space out	Proper space out of assemblies Tally Prepare and double checked	Rigging Up	Missalignment between drilling depth and completion tally.	Excel errors, measurement equipment's, make up loss error.		Zonal isolation compromised	Follow the operations program to make the tally Communication with the involved personnel
3.2	W/U system on rig floor	Rig crew slinging the WEA-ISP for pick up to the rig floor/cat walk	Rigging Up	Wrong placement of the sling on WEA-ISP.	Lack of communication between WT and rig crew Adverse weather, lifting operations pose danger	WEA-ISP damage and seal damage	Delay of work, potential HPT Replacement of equipment	QA/QC of the Tally with all parties review and acceptance Welltec engineer to supervise the slinging process to make sure they put the sling on the correct place. Rig crew to have training and awareness on the weather condition effect on the lifting process.
3.3	W/U system on rig floor	Make up WEA-ISP assembly to the completions string	Rigging Up	Unable to make the thread connection.	Not centralization, Not walking three d.in.	Damaged thread or connection	Delay of work, potential HPT Replacement of equipment	Make sure rig crew uses the correct tool for making up such as Stabbing Guide. Having back up equipment on location
3.4	RIH with system and fill with wellbore fluid	Running in the hole	Deployment	Damage to the WEA-ISP's sleeve	Passing through the rotary table with smaller ID than the WEA-ISP OD	WEA-ISP sleeve or seals damaged	Loss of zonal isolation. Efficiency of stimulation/production affected	Check and measure the dimensions of the WEA-ISP to run in hole to see compatibility with rotary table during the planning of the operation. Prepare the rig crew for any changes needed to avoid lost time
3.5	RIH with system and fill with wellbore fluid	Running in the hole	Deployment	Damage to the WEA-ISP's sleeve	Running in hole at high speed across the previous liner top (sharp edges)	WEA-ISP damaged while running in hole	Loss of zonal isolation. Efficiency of stimulation/production affected	Include maximum running speed in the program. Toolbox talk on the running procedures. Welltec to provide information
3.6	RIH with system and fill with wellbore fluid	Running in the hole	Deployment	Damage to WEA-ISP seal stack	High pumping rate (job specific - e.g. >25bbl for a 8.25" drift OD in a 9" hole)	Damage to seals in the WEA-ISP sleeve	Degradation of zonal isolation.	Include maximum pumping rate in the program. Toolbox talk on the running procedures. Welltec to provide information Redundant seal system Slim running CO WEA-ISP tested and qualified to provide effective isolation in scenario where seals are damaged
3.7	Pressure up WEA-ISP to the desired pressure.	Expanding WEA-ISP	WEA-ISP Expansion	Not able to get to desired pressure. Unable to set WEA-ISP packers	Possible leak in surface equipment Possible leak in downhole equipment Possible leak of WEA-ISP downhole		Lost time to get replacement equipment or fix connection Lost time to find the find the leak Lost time to find the leak and set the WEA-ISP using contingency plan	Test surface lines prior to pressure up the whole completions string. Contingency equipment for WEA-ISP setting such as cup tool Interaction with for leak detection mitigation
3.8	Hold the desired pressure according to WEA-ISP setting procedure (5-10 minutes)	Expanding WEA-ISP	WEA-ISP Expansion	Not fully engage/expand WEA-ISP Over pressure of WEA-ISP	Not following procedure Density difference between annulus and bore, surface pressure equipment issue	Not all seals engage and energize to the casing Exceed force and stress to casing	Degradation of zonal isolation.	Follow operational program and perform calculations of downhole pressures prior to running in the well Downhole gauge to show differential pressure on setting the WEA-ISP.
3.9	Integration of XTR packer and WEA-ISP packers and release setting tool	Setting XTR and WEA-ISP packers and release setting tool	WEA-ISP Expansion XTR Packer setting	Unable to set WEA-ISP packers XTR Packer does not set	Issue with expansion sleeve Compatibility of setting pressures Setting differences between WEA-ISP and XTR	Not able to seal in the desired zones Unable to operate XTR or WEA-ISP systems	Loss of zonal isolation. Efficiency of stimulation/production affected	Setting procedures for WEA-ISP packer - shop test to ensure it integrates with XTR packer Ensure compatibility of setting pressures in downhole conditions with XTR pack WEA-ISP
3.1	Impact to space out due to longer WEA-ISP than iso packers (need to compare difference in length)	Job Planning	Deployment	Iso packers with shorter dimensions	Space out of system not compatible	Unable to deploy the system for effective isolation		WEA-ISP can be manufactured to different lengths, follow proper job planning and equipment planning

Figure 13—FMECA example section of the Pick-Up(PU) and Run-in-hole (RIH) with MSST System

The evaluation concluded that the WEA-ISP could be incorporated into the MSST system without requiring significant modifications to the completion design or execution procedures.

Job Planning. Pre-job planning sessions were held with representatives from the operator, MSST provider, TCP team, and WEA-ISP engineering group. These meetings focused on:

- Final review of well and equipment specifications.
- Verification of tool functionality and operational limitations
- Confirmation of Standard Operating Procedures (SOPs)

All WEA-ISP components were inspected and verified prior to assembly. Equipment was shipped to the MSST provider's workshop, where bucking operations were performed to integrate the WEA-ISP into the MSST string. The final completion tally and survey for the trial well were reviewed and approved prior to deployment (Figure 14).

Operation	Lib	Description	Acq. Len	Bit Depth	Row	Pin	Basket
1	1	1st Hoisting Joint w/ Midhook	No Drift	43.42	43.42	5-1/2" VAM	Midhook
2	1	2nd Hoisting Joint	No Drift	41.87	85.09	5-1/2" VAM	5
3	1	3rd Hoisting Joint	No Drift	43.27	128.36	5-1/2" VAM	5
4	1	4th Hoisting Joint	No Drift	20.42	148.78	5-1/2" VAM	7
5	1	PCS Chase Shifters	No Drift	40.88	40.88	4" Hpl 511	Midhook
6	1	Wrenching It w/ 8" Pipe	No Drift	37.83	78.71	4" Hpl 511	4" Hpl 511
7	1	Lower Gauge Center Assy	No Drift	34.39	113.04	4" Hpl 511	5
8	1	Midhook Gauge Center Assy	No Drift	34.22	147.26	4" Hpl 511	4" Hpl 511
9	1	Upper Gauge Center Assy	No Drift	35.88	182.94	4" Hpl 511	5
10	1	WCS Shifters Assy w/ 8" Pipe (8" Pipe Handy)	No Drift	57.14	240.07	4" Hpl 511	4" Hpl 511
11	1	Service Tool	No Drift	51.08	291.15	4" Hpl 511	5
12	1	Space Out Joint	No Drift	31.52	322.67	4" Hpl 511	5
13	1	Space Out Joint (8 Strands in Derrick)	Drifted	742.23	1,064.90	4" Hpl 511	Derrick
14	1	Primary Intermediate Seals	No Drift	89.12	1,154.02	4" Hpl 511	5
15	1	Primary Test Pops	No Drift	17.92	1,171.94	4" Hpl 511	5
16	1	1st Sump Pack Assy	No Drift	124.46	1,296.40	4" Hpl 511	5
17	1	Zone 1 Sump Seal Assy (8" 11.88 S11 HFD-Stack Outlets Drift)	No Drift	58.40	1,354.80	5-5/8" VAM	5-5/8" VAM
18	1	Zone 1 Production Joint	NO DRIFT	57.86	1,412.76	5-5/8" VAM	5-5/8" VAM
19	1	Zone 1 Mid Joint	NO DRIFT	58.24	1,471.00	5-5/8" VAM	5-5/8" VAM
20	1	Zone 1 Top Joint	NO DRIFT	21.47	1,502.47	5-5/8" AB-HDL	5-5/8" VAM
21	1	Zone 1 Blank Assy w/ Adapter	Test	55.37	1,557.84	5-5/8" AB-HDL	5-5/8" AB-HDL
22	1	Zone 1 Grand Pack Assy	Test	55.37	1,613.21	5-5/8" AB-HDL	5-5/8" AB-HDL
23	2	Zone 2 Half Joint Screen & WAB Packer	NO DRIFT	65.47	1,678.68	5-5/8" VAM	5-5/8" VAM
24	2	Zone 2 Production Joint	NO DRIFT	65.22	1,743.90	5-5/8" VAM	5-5/8" VAM
25	2	Zone 2 Mid Joint Assy	NO DRIFT	54.95	1,798.85	5-5/8" VAM	5-5/8" VAM
26	2	Zone 2 Top Joint	NO DRIFT	58.20	1,857.05	5-5/8" VAM	5-5/8" VAM
27	2	Zone 2 Blank Assembly w/ Adapter	NO DRIFT	21.39	1,878.44	5-5/8" AB-HDL	5-5/8" VAM
28	2	Zone 2 Grand Pack Assy	NO DRIFT	56.15	1,934.59	5-5/8" VAM	5-5/8" AB-HDL
29	2	Zone 3 Half Joint Screen & WAB Packer	NO DRIFT	65.41	1,999.99	5-5/8" VAM	5-5/8" VAM
30	2	Zone 3 Production Joint	NO DRIFT	65.27	2,065.26	5-5/8" VAM	5-5/8" VAM
31	2	Zone 3 Mid Joint Assy	NO DRIFT	54.95	2,120.21	5-5/8" VAM	5-5/8" VAM
32	2	Zone 3 Top Joint	NO DRIFT	58.26	2,178.47	5-5/8" VAM	5-5/8" VAM
33	2	Zone 3 Blank Assembly w/ Adapter	NO DRIFT	21.51	2,239.98	5-5/8" AB-HDL	5-5/8" VAM
34	2	Zone 3 Grand Pack Assy	NO DRIFT	56.21	2,296.19	5-5/8" VAM	5-5/8" AB-HDL
35	2	Zone 4 Half Joint Screen & WAB Packer	NO DRIFT	65.40	2,361.59	5-5/8" VAM	5-5/8" VAM
36	2	Zone 4 Production Joint	NO DRIFT	65.25	2,426.84	5-5/8" VAM	5-5/8" VAM
37	2	Zone 4 Mid Joint Assy	NO DRIFT	54.94	2,481.78	5-5/8" VAM	5-5/8" VAM
38	2	Zone 4 Top Joint	NO DRIFT	58.26	2,540.04	5-5/8" VAM	5-5/8" VAM
39	2	Zone 4 Blank Assembly w/ Adapter	NO DRIFT	21.49	2,601.53	5-5/8" AB-HDL	5-5/8" VAM
40	2	Zone 4 Grand Pack Assy	NO DRIFT	56.14	2,657.67	5-5/8" VAM	5-5/8" AB-HDL
41	2	Zone 4 Lower MCS Assy	Test	10.81	2,768.48	HAL	5-5/8" VAM
42	2	Zone 4 Fishing String	No Drift	12.89	2,881.37	4" Hpl 511	Ball Valve
43	2	Zone 4 Fishing String (8 Strands in Derrick)	Drifted	893.29	3,774.66	4" Hpl 511	Derrick
44	2	WAB Packer (8" 11.88 S11 HFD-Stack Out)	No Drift	55.07	3,829.73	CMST	HAL

Figure 14—Tally of the completion string with the WEA-ISP integrated in the system

Job Execution. The MSST completion followed standard operational procedures:

1. **Sump Packer Deployment** - The sump packer was set as the initial step in the completion sequence.
2. **TCP Run and Perforation** - Tubing Conveyed Perforating (TCP) was executed to perforate the designated zones.
3. **MSST Deployment Without Deburring** - Immediately following the TCP run and retrieval of the guns, the MSST system was deployed without performing a deburring run. This marked a significant departure from conventional practice.

At the rig site, close coordination among the MSST provider, operator, and WEA-ISP personnel ensured smooth execution. All RIH parameters and limitations were communicated clearly, and the MSST system was deployed successfully without encountering issues related to burr interference.

WEA-ISP Expansion and Activation. Once the MSST system was stung into the sump packer, the pressure cycles for setting the packers were initiated:

- **Production/Injection Packer Setting** - The initial pressure cycle reached 2,500 psi to activate the conventional Production/Injection packer at the top of the MSST assembly (Figure 15).
- **WEA-ISP Expansion** - A subsequent pressure cycle was applied to expand all WEA-ISP units (spaced between perforated intervals) simultaneously. The system was pressurized to 5,000 psi and held for the required duration to ensure full expansion and sealing (Figure 16).

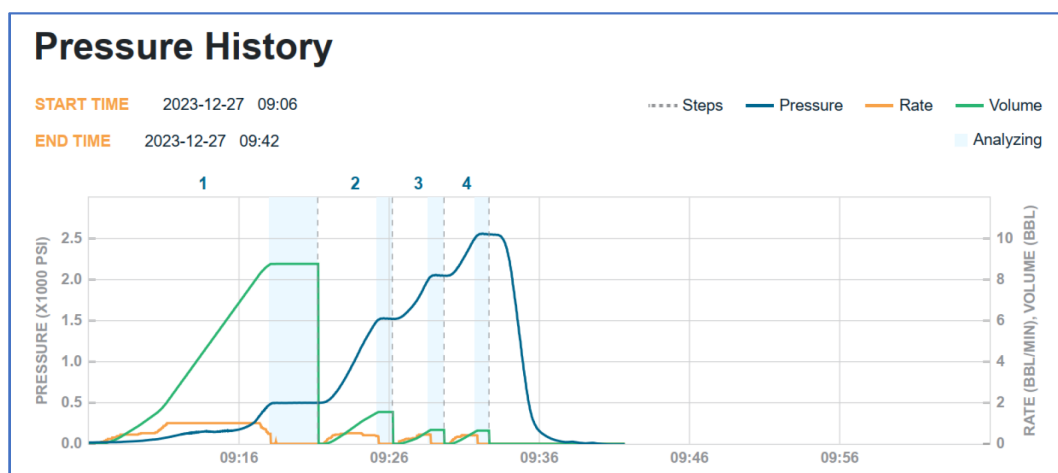


Figure 15—Pressure cycle to 2,500 psi to set Production/Injection packer

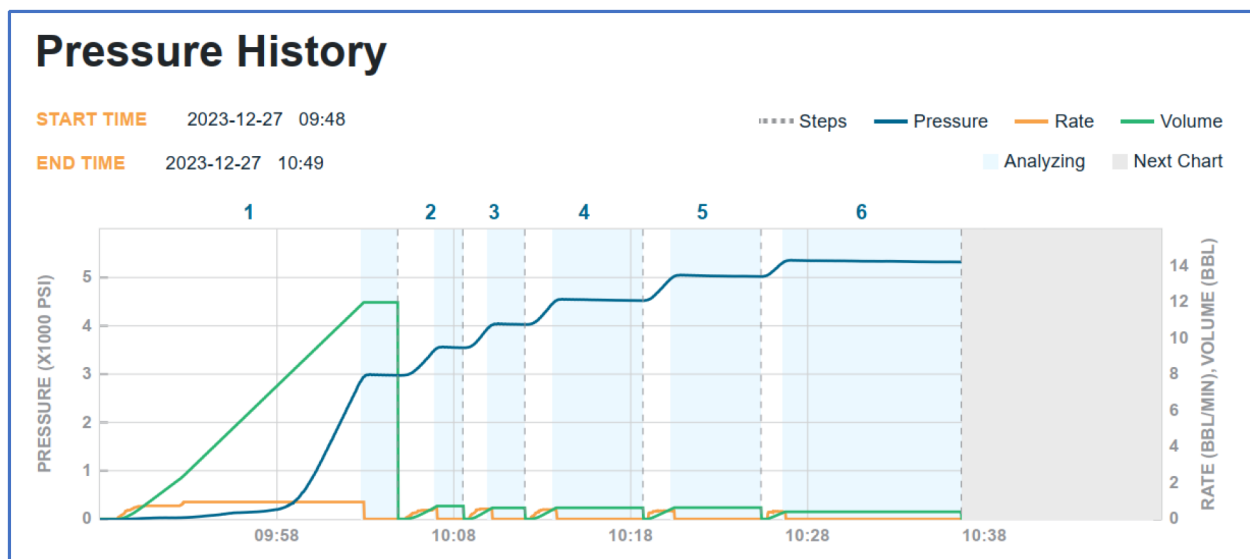


Figure 16—Pressure cycle to 5,000 psi to set all the WEA-ISP simultaneously

Both pressure cycles were executed successfully, confirming the operational readiness and reliability of the WEA-ISP system under field conditions.

Results and observations

The deployment of three WEA-ISP units in a single MSST completion, *executed without a deburring run*, successfully achieved zonal isolation and enabled frac-and-pack stimulation across all targeted intervals. The well was completed without operational complications, validating the system's integrity and performance under field conditions. This approach resulted in a rig time savings of almost three days, directly attributable to the elimination of the deburring operation.

Following the initial success, three additional deepwater wells have been completed over an 18-month period using the same methodology. Across the four wells, the operator would have realized a cumulative savings of \$11MM, based on an assumed spread rate of \$1MM per day. These savings were achieved by eliminating four deburring runs, (one per well) without compromising completion quality or well integrity.

The WEA-ISP system demonstrated consistent performance, with its slim outer diameter facilitating smooth run-in-hole operations and reliable expansion. The system's ability to withstand burr-induced damage and maintain sealing and anchoring integrity was confirmed through both laboratory qualification and field execution.

The [table 3](#) shown above demonstrates the efficiency gain of days (~11 days) over the last four well sin the project, significantly reducing operational and occupational risks presented during the deburring operation.

Table 3—Operational Cost Comparison Across Four Wells

Step	Task Description	With Deburring Run (4 Wells)	Without Deburring Run (4 Wells)
1	Set sump packer via wireline	2 days	2 days
2	Perforate well using TCP	10 days	10 days
3	Deburr perforations	11 days	0 days
4	Deploy MSST and seal	8 days	8 days
	Total Rig Time	30.8 days	20 days
	Total Cost	\$32M USD	\$20M USD

Disclaimer

Any comments regarding the co-author Welltec, should be used for informational purposes and should not imply an endorsement of any kind.

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Nomenclature

ACS	= Annular Cementing Sleeve
API	= American Petroleum Institute
DNV	= Det Norske Veritas
ECD	= Equivalent Circulation Density
FAT	= Factory Acceptance Test
FEA	= Finite Elemental Analysis
GoA	= Gulf of America
JRA	= Job Risk Assessment
MEA	= Metal Expandable Anchor
MEP	= Metal Expandable Packer
NPV	= Net Present Value
RIH	= Rin In Hole
SOP	= Standard Operating Procedure
TCP	= Tubing Conveyed Perforating

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